Rajaee Mundle

University of Guelph | Mechanical Engineering (B.Eng.) | Graduated June 2026

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Skills Summary

CAD & Simulation: SolidWorks (CSWA) | AutoCAD | Fusion 360 | OnShape | FEA

Design & Manufacturing: GD&T | DFM | DFMA | Composite Manufacturing | Lean Manufacturing | Tooling & Fixtures

Tools & Automation: Drill Press | Lathe | Mill | Laser Cutting | Robotic Welding Systems (Fanuc/Panasonic) | 3D Printing

Experience

Manufacturing Engineer Intern Jan 2024 – Aug 2024

SAF-HOLLAND *Woodstock, ON*

- Designed and executed 12+ custom manufacturing fixtures from concept to production, including a spacer weld fixture that utilized a fixed-point datum to align axles by part number precisely.
- Engineered a multipurpose, modular weld table for a Panasonic robotic weld cell, designing 3 interchangeable fixtures to accommodate multiple plant components and their geometric variations.
- Developed and modelled 3 landing gear weld fixtures and a specialized trailer foot fixture to streamline plant assembly processes.
- Spearheaded the physical overhaul of legacy hydraulic presses, integrating modern safety features and robust structural enhancements to transform them into high-capacity, heavy-duty machinery.
- Optimized a 16 ft × 16 ft weld cell layout using AutoCAD, organizing new welding gear, protective curtains, and a separation barrier for the sandblaster to maximize floor space efficiency.
- Partnered with a team of 6 millwrights to facilitate hands-on fabrication, installation, and real-time troubleshooting of newly designed shop fixtures.
- Conducted cycle-time optimization studies on manual loading stations and introduced design updates to minimize structural rework on defective axles.

Engineering Club Experience

Aerodynamics Co-Lead|Vehicle Engineer Sept 2021 – Jun 2026

Guelph Gryphon Formula SAE team *Guelph, ON*

- Collaborated with senior team members to learn vehicle integration, design workflows, and manufacturing processes used in racecar development.
- Supported vehicle manufacturing and assembly, specifically structural mounts.
- Assisted with composite mould preparation and carbon fibre part fabrication for Formula SAE aerodynamic packages.
- Managed a 40+ member aerodynamics team, coordinating off-site developmental meetings and structuring timelines to deliver 3 aerodynamic kit designs.
- Led hands-on composite manufacturing, including the physical tooling, layup, and structural assembly of carbon fibre nosecones, sidepods, and wings.
- Owned the geometric design of vehicle sidepods and radiator shrouds, ensuring integration with powertrain cooling requirements while passing all formal technical inspections.

Projects

Welding Fixture | *SolidWorks, Tooling & Fixtures, Automated Weld Cells* Design & Manufacturing Project

- Reverse-engineered a heavy-duty production welding fixture for trailer landing gear feet, eliminating 100% of manual setup variability and dimensional alignment deviations.
- Engineered robust locating features and repeatable clamping mechanisms in SolidWorks to maintain tight machining tolerances within ±0.075in.

Scale-Model Compactor Bin | *SolidWorks, Arduino IDE, 3D printing* Lab Project

- Engineered an automated, sensor-activated household waste compaction prototype utilizing an Arduino Uno R3, an HC-SR04 ultrasonic sensor, and dual Miuzei MG90S servo motors for fully touchless operation.
- Modelled a 3-part structural enclosure assembly (base, head housing, and inner can) in SolidWorks, managing dynamic clearance constraints to prevent compression plate binding.
- Conducted a static structural Finite Element Analysis (FEA) simulation applying a 900N distributed load to evaluate maximum von Mises stress levels (304.2 MPa localized stress concentrations) and validate PLA structural integrity.
- Integrated a 12V DC power distribution system featuring an 18 AWG safety fuse, a BTS7960 high-current H-bridge motor driver, and a 12V-to-5V buck converter for electronic signal regulation.
- Fabricated all custom structural elements using additive manufacturing (PLA 3D printing) and executed physical testing under load, proving structural stability and crushing aluminum cans.